An Ultra-Low-Power SAR ADC with an Area-Efficient DAC Architecture

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***Abstract*—An ultra-low-power area-efficient 8-bit successive approximation register (SAR) analog-to-digital converter (ADC) is presented. To achieve ultra-low-power performance a DAC architecture is proposed that employs two rail-to-rail low-power unity-gain buffers and only 4 minimum-size capacitors instead of the conventional binary-weighted capacitor array. Thereby, power consumption and area are drastically reduced by virtue of lower switching activity and smaller size capacitor array. The proposed 8-bit SAR ADC is designed and simulated in a 0.13μm CMOS process. Simulation results show that for a 2.4 kHz (12.4 kHz) input signal while sampling at 25 kHz, the ADC achieves an ENOB of 7.9 (7.8), consumes 290 nW (350 nW) form a 0.8 V analog supply and a 0.6 V digital supply, and achieves a FoM of 48 fJ/conversion-step (62 fJ/conversion-step).**

I. Introduction

Emerging mixed-signal applications such as radio-frequency identification (RFID) systems, wireless sensor networks, portable biometric devices and energy scavenging systems demand a high power efficiency to secure a long autonomous operation. Analog-to-digital converters (ADCs) are among critical components that are used in most of such systems and usually consume a fair share of the overall power budget. Among different ADC architectures, successive approximation register (SAR) ADCs [1-6] are the most popular candidate for the aforementioned applications which typically require high-resolution, low-to-medium speed and low power.

Conventional SAR ADCs use the standard binary search technique [1-6]. Generally, in a SAR ADC there are two major contributors to power consumption: comparator and charging/discharging of the DAC array. However, comparator's power consumption is usually not a matter of concern unless a very high-speed and high resolution is required. (Yet, a comparator architecture compatible with the proposed digital-to-analog converter (DAC) structure is briefly discussed in the paper.)

Typical SAR structures use bulky DAC capacitive arrays which are usually the area and power bottleneck of the design. From the power dissipation perspective, in order to sample a full-scale sine wave at the Nyquist frequency, a traditional capacitive array on average consumes [7]:

$$P_{\text{in}} = \frac{C_{\text{T}} V_{\text{R}}^2}{T_{\text{s}}} \tag{1}$$

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where P_{in} is the power drawn from the input, C_{T} is the total capacitance of the array, V_{R} is the reference voltage and T_{s} is the conversion period. The average power drawn form the reference generator for charge redistribution is [7] :

$$P_{\text{ref}} = \frac{C_{\text{T}} V_{\text{R}}^2}{2T_{\text{s}}} \tag{2}$$

Accordingly, layout area and power consumption grow exponentially ($C_{\text{T}} = 2^n C_0$) with the number of bits (n) which compromises the advantage of SAR ADCs in terms of area and power, especially as the number of bits increases.

Extensive studies have been carried out on energy-efficient DAC structures [1-8]. [2] proposes the capacitor splitting scheme which reduces the average switching energy of the array by splitting the most significant bit (MSB) capacitor into $n-1$ binary scaled sub-capacitors (where n is the number of bits). Consequently, for down transitions [2], the charge on the MSB capacitor will be restored for further conversions. Although the proposed technique significantly saves the switching energy for small output codes, it fails to perform equally well for higher output codes where less down transitions occur. Moreover, this approach is still dependant on a bulky DAC array and requires twice as many switches as the conventional scheme.

In another structure, [8] reduces the average switching energy of the capacitive array by separating the decoding of MSB and least significant bit (LSB) through using two different capacitor arrays with unequal size. However, the low-power consumption is achieved at the cost of two additional clock cycles which poses energy overhead due to additional comparison steps required. Moreover, the layout occupies a higher area than that of the traditional architecture due to MSB DAC, memory storage with logic and the extra comparator.

This paper proposes an area-efficient DAC architecture that both reduces the power consumption and the layout area. The proposed DAC architecture consists of four minimum-size capacitors and two low-power buffers for which the binary search algorithm is carried out at no extra cost in terms of clock cycles and comparison steps. Also, the use of four minimum-sized capacitor DAC relaxes capacitor mismatch and parasitic requirements. To further save area and power consumption, the two low-power buffers are dually used as the comparator preamplifier blocks in an open-loop fashion. Section II presents the DAC architecture while Section III describes the proposed comparator. In Section IV, the sample-

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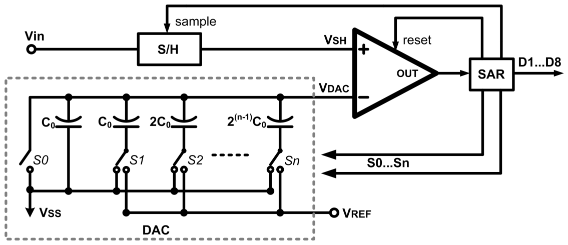

Fig. 1. Conventional SAR architecture

and-hold architecture is described, and Section V provides the simulation results.

II. DAC Architecture

The operating principles of a SAR converter are well known [1-7] . Simply put, the capacitive DAC main task is to provide the appropriate voltage levels to implement the binary search algorithm as shown in Fig. 1. The algorithm starts by setting the MSB to '1' and all other bits to '0'. Thereby, V_{in} is compared with $\frac{V_{\text{ref}}}{2}$. Charge recycling is then carried out based on the output of the comparator, i.e., if D_1 (comparator output) is high, the second largest capacitor is charged to V_{ref} setting $V_{\text{DAC}} = \frac{3V_{\text{ref}}}{4}$. This is called an "up" transition. Conversely, if D_1 is low, the largest capacitor is discharged and the second largest one is charged to V_{ref} which is called a "down" transition. The sequence continues until all the bits are decided. The binary search algorithm described above can be interpreted as follows: two voltage levels are defined as V_{top} and V_{bot} which are the upper and lower voltage bounds at each clock cycle and they converge toward V_{in} with each bit decision. V_{mid} is then produced from V_{top} and V_{bot} as:

$$V_{\text{mid}} = \frac{V_{\text{top}} + V_{\text{bot}}}{2} \tag{3}$$

For the m^{th} bit, this voltage, i.e., the output of the DAC, is compared with V_{in} and the comparison result D_m updates the value of V_{top} or V_{bot} in the next clock cycle such that if $\text{D}_\text{m} = 1$ ('up' transition), V_{top} maintains its old value while V_{bot} is shifted up to V_{mid}. Conversely, if $\text{D}_\text{m} = 0$ ('down' transition), V_{bot} retains the old value and V_{top} is shifted down to V_{mid}. The procedure continues until all bits are concluded. As shown in Fig. 2 the algorithm can be implemented using four capacitors and two unity gain buffers. For a given output of '001...' (where the first three bits are evaluated) the operation of the proposed DAC architecture can be described as follows: Each bit decision cycle is divided into two time slots, comparison cycle and charge recycling cycle. During each comparison cycle one capacitor set $(\text{C}_{\text{topi}}, \text{C}_{\text{boti}})$ performs as a capacitive divider producing $V_{\text{mid}} = \frac{V_{\text{top}}+V_{\text{bot}}}{2}$ and the other set holds the same two voltage levels $(V_{\text{top}}, V_{\text{bot}})$ that are being processed (the two capacitor sets interchangeably switch these two tasks in subsequent comparison cycles). Voltage levels are then updated at charge recycling cycle based on the previous decision. At the beginning of conversion (sampling phase), (S_1, S_7) and (S_2, S_8) are on, charging $(\text{C}_{\text{top1}}, \text{C}_{\text{top2}})$ and $(\text{C}_{\text{bot1}}, \text{C}_{\text{bot2}})$ to V_{REF}

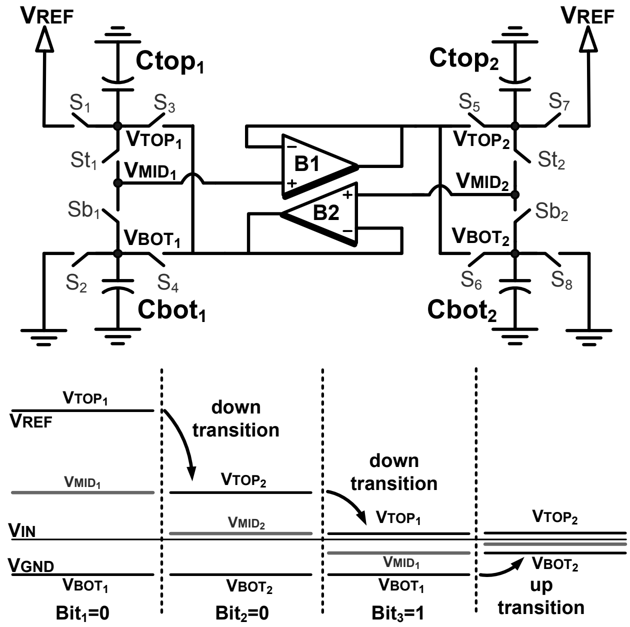

Fig. 2. New DAC architecture and timing diagram.

and V_{GND}, respectively. Then in the first comparison cycle, (S_1, S_2) and (S_7, S_8) turn off and $(\text{S}_{\text{t1}}, \text{S}_{\text{b1}})$ turn on to form a capacitive divider (from $\text{C}_{\text{top1}}, \text{C}_{\text{bot1}}$) which gives $V_{\text{MID1}} = \frac{V_{\text{REF}}}{2}$. Note that as stated earlier when each capacitor set is active, i.e., providing V_{MID} for comparison $(\text{C}_{\text{top1}}, \text{C}_{\text{bot1}})$, it is essential that the other capacitor set $(\text{C}_{\text{top2}}, \text{C}_{\text{bot2}})$ holds the same voltage levels being searched at the time, which in this case is $V_{\text{TOP2}} = V_{\text{REF}}$ and $V_{\text{BOT2}} = 0$. Then V_{MID1} is compared with V_{IN} which yields $\text{D}_1 = 0$ based on the assumption made for the first three bits. Voltage levels are updated in the succeeding charge recycling cycle. As mentioned earlier for each 'down' transition V_{bot} maintains its value and V_{top} is shifted down to V_{MID}. For this purpose, (S_{t1}, S_{b2}) and (S_4, S_5) are switched on while all other switches are off. Buffer B_1 provides a copy of V_{MID1} for C_{top2} thus shifting V_{top2} down to V_{MID1}. At the same time, buffer B_2 returns C_{bot1} voltage back to $V_{\text{BOT1}} = V_{\text{GND}}$ in order for $(\text{C}_{\text{top1}}, \text{C}_{\text{bot1}})$ to have a copy of new voltage levels that are to be processed in the next comparison cycle. At this point, $V_{\text{top1}} = V_{\text{top2}} = \frac{V_{\text{REF}}}{2}$ and $V_{\text{bot1}} = V_{\text{bot2}} = 0$ and the capacitors are all set for the next comparison cycle where (S_{t2}, S_{b2}) turn on and all other switches are off. The capacitive divider produces $V_{\text{MID2}} = \frac{V_{\text{TOP2}}+V_{\text{BOT2}}}{2} = \frac{V_{\text{REF}}}{4}$. This voltage is compared with V_{in} and results in $\text{D}_2 = 0$ which corresponds to a 'down' transition. In the next charge recycling cycle, (S_{t2}, S_{b1}) and (S_3, S_6) are switched on. Buffer B_2 shifts V_{TOP1} down to $V_{\text{MID2}} = \frac{V_{\text{REF}}}{4}$ and buffer B_1 returns V_{BOT2} back to V_{GND} such that $V_{\text{top1}} = V_{\text{top2}} = \frac{V_{\text{REF}}}{4}$ and $V_{\text{bot1}} = V_{\text{bot2}} = 0$. Therefore, in the next comparison $(C_{\text{top1}}, C_{\text{bot1}})$ produce $V_{\text{MID1}} = \frac{V_{\text{REF}}}{8}$ which yields $\text{D}_3 = 1$. In the next charge recycling cycle (S_{t2}, S_{b1}) and (S_3, S_6) are switched on. Buffer B_1 pulls V_{BOT2} up to $V_{\text{MID1}} = \frac{V_{\text{REF}}}{4}$ and buffer B_2 returns V_{TOP1} back to $\frac{V_{\text{REF}}}{2}$. All the next bits are decided in the same manner.

In terms of power, the proposed architecture outperforms

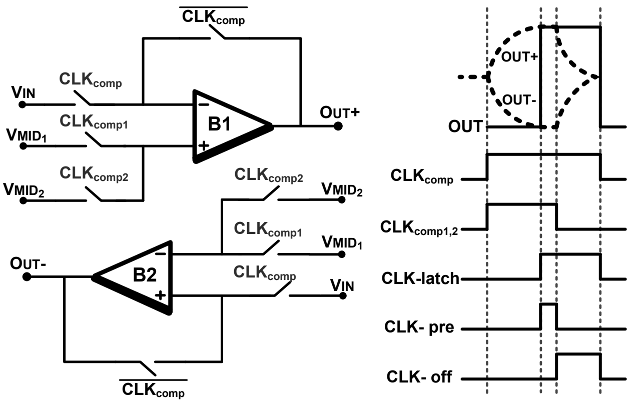

Fig. 3. Proposed comparator architecture and timing diagram.

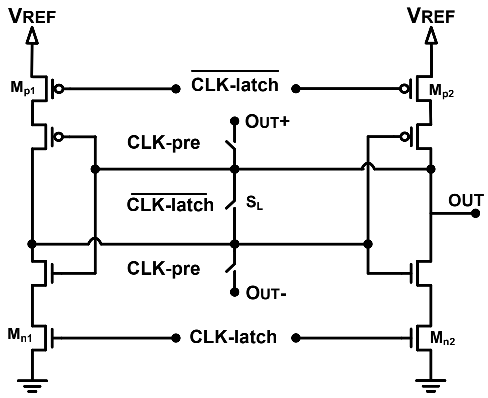

Fig. 4. Low-power latch architecture.

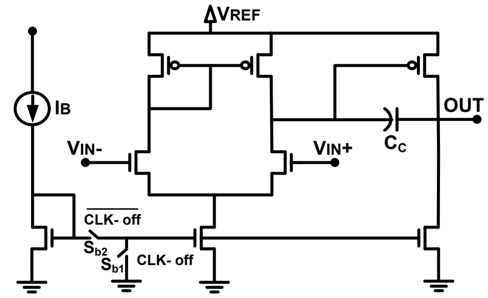

Fig. 5. Low-power amplifier.

the conventional SAR architectures. As the algorithm suggests, during a conversion cycle, each of the two buffers contribute in the charge recycling process for at most n times. So, the total charge drawn from V_{REF} during the conversion cycle is:

$$Q_{\text{tot}} = 4 \cdot n \cdot I_{\text{b}} \cdot T_{\text{CR}} \tag{4}$$

where n is the number of bits, I_{b} is the bias current for each buffer stage, T_{CR} is the charge recycling period and the factor '4' accounts for the two, two-stage amplifiers. the two factors governing I_{b} are the required voltage gain and speed. To achieve the desired resolution, the buffer output has to settle within $\frac{1}{2}$LSB of the input, which for an 8-bit resolution requires an open loop gain on the order of 50 to 60 dB. The two-stage low-power amplifier (described in Section III) provides such a gain by drawing only 124 nW from V_{REF}. Regarding speed, since the load capacitance is minimum-sized, slew-rate is not that critical and charge-recycling period (T_{CR}) can be very short.

The aforementioned algorithm is implemented in the proposed ADC with some minor amendments. To further save power, the buffers turn off whenever their service is not required. This is achieved by shutting off the supply current (the buffer architecture is described in Section III). An example for such a power saving mode is when the destination voltage, V_{TOP} or V_{BOT} is supposed to be updated to V_{REF} or V_{GND}, respectively. In this situation the respective buffer will not perform the task, instead one of (S_1, S_7) or (S_2, S_8) turns on to update V_{TOP} or V_{BOT} to V_{REF} or V_{GND}, respectively. This power saving scheme proves useful particularly for very large or very small inputs. For instance, for a very small output code starting with several consecutive '0's (0000...), at each charge recycling cycle, one buffer remains inactive until the first '1' is detected. The same goes for very large output codes.

III. Comparator Architecture

As mentioned in Section I, to further save area and power, the two buffers employed in DAC are used as the preamplifier stages for the comparator by opening the loop through $\overline{\text{CLK}_{\text{comp}}}$, as shown in Fig. 3. At each comparison phase, one capacitor set provides V_{MID} to be compared with V_{IN} for which the appropriate clock ($\text{CLK}_{\text{comp1}}$ or $\text{CLK}_{\text{comp2}}$) goes high. To better utilize the available hardware, the two amplifiers operate simultaneously to amplify the voltage difference between the input voltages. For this purpose V_{IN} and V_{MIDi} are applied to opposite terminals of B_1 and B_2. Thus, as one of OUT+/OUT- charges up to V_{REF}, the other one is pulled down to V_{GND}. This scheme brings a twofold benefit: on the one hand, the pre-amplification phase would be twice as fast. On the other hand the overall comparator resolution is improved.

Almost at the end of comparison cycle where OUT+ and OUT- are about their extreme voltage levels (note that clock periods are not drawn to scale in Fig. 3), a very short pulse CLK_{pre}, connects OUT+ and OUT- to the input terminals of the regenerative latch as shown in Fig. 4. CLK_{pre} turns off after a short period so that B_1 and B_2 do not drive the latch for the rest of the latch cycle. The buffer/amplifier architecture is shown in Fig. 5. After the pre-amplified voltages are properly delivered to the latch at the end of CLK_{pre}, CLK_{off} goes high which turns the amplifiers off by shutting off the bias currents through $\text{S}_{\text{b1,2}}$.

Latch phase starts simultaneously with CLK_{pre} and the regenerative latch produces a digital level output in a very short interval. At the end of process, S_{L} resets the latch by connecting the input and output of the back-to-back inverters to set them close to $\frac{V_{\text{REF}}}{2}$ which speeds up the next latch operation. To further save power, $\text{M}_{\text{n1,2}}$ and $\text{M}_{\text{p1,2}}$ perform the power-gating task by blocking the latch input current in inactive mode when $\text{CLK}_{\text{latch}}$ is off.

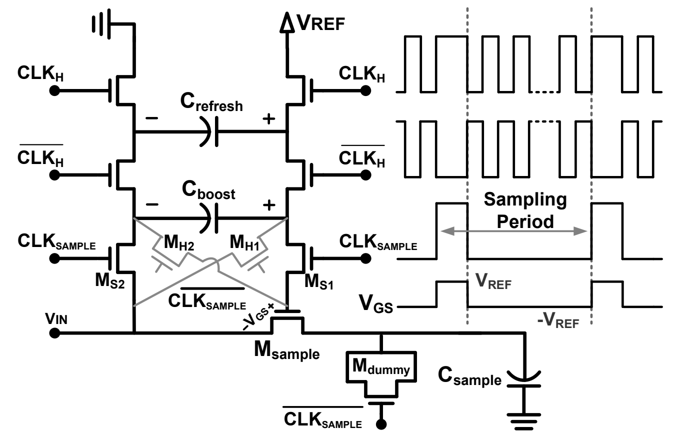

Fig. 6. Proposed Sample-and-Hold architecture and timing diagram.

IV. Sample and Hold Architecture

A switched-capacitor clock-boosted architecture is used for the sample-and-hold circuit due to its rail-to-rail capability and low-power operation. As shown in Fig. 6, after a few clock periods, C_{boost} is charged to V_{REF} and subsequently acts as a floating voltage source. At each sampling phase (CLK_{SAMPLE}), C_{boost} connects the gate-source terminals of M_{sample} in such a way that: $V_{GS} = V_{REF}$. Therefore in sampling phase, the gate voltage of M_{sample} will always be as much as V_{REF} higher than its source voltage, regardless of the value of V_{IN}. This presents a rail-to-rail operation. On the other hand, in the hold phase, M_{H1}, M_{H2} turn on and M_{S1}, M_{S2} turn off, providing a negative voltage between the gate-source terminals of M_{sample} such that for the entire hold phase: $V_{GS} = -V_{REF}$. This negative bias alleviates the leakage current and fully isolates the sampled voltage from input fluctuations.

CLK_H can be picked from any of the clock signals already generated in the circuit, however, for a better performance, it has to be a high frequency clock such as CLK_{comp}. As shown in Fig. 6, $C_{refresh}$ updates the voltage on C_{boost} at every $\overline{CLK_H}$. M_{dummy} is employed to cancel the charge-injection effect. However, special care has to be given to the size of M_{dummy} which is not the conventional $\frac{Size_{MSample}}{2}$ due to the fact that the gate-source voltages that the two transistors experience are not the same. Note that this architecture almost imposes no power consumption penalty as the current drawn from V_{REF} is limited to the leakage current of C_{boost} and switching transistors and thus is on the order of few nano Watts.

V. Simulation Results

The proposed DAC/comparator architecture is designed and laid out in an 8-bit 25kS/s SAR ADC in a 0.13μm CMOS technology. DAC capacitors are implemented by a MIM structure with the unit capacitor of 50 fF and the sampling capacitor is 500 fF. As shown in Fig. 7, at 25kS/s the simulated DNL and INL are within 0.25 LSB and 0.5 LSB, respectively. The spectra for an input full-scale sine wave at 2.24 kHz and 12.4 kHz are shown in Fig. 8, for which the SNDRs of 49.4 dB (ENOB=7.91) and 48.8 dB (ENOB=7.81) are obtained, respectively.

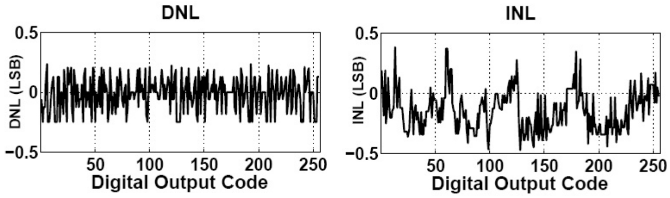

Fig. 7. Simulated DNL and INL at 25kS/s.

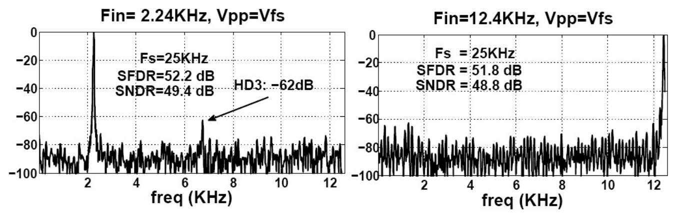

Fig. 8. Output spectrums for input sine wave @ 2.24kHz and 12.4kHz.

The entire digital circuitry is custom designed and dissipates 24 nW from a 0.6 V supply. The total power consumption for 2.24 kHz and 12.4 kHz input sine-wave are simulated to be 290 nW and 350 nW which exhibits a FoM of 48 fJ/conversion-step and 62 fJ/conversion-step, respectively. A comparison of the proposed architecture with the state-of-the-art SAR ADCs is provided in Table I.

TABLE I
Comparison of the Proposed SAR ADC with the State-of-the-Art Results

	[1]*	[3]*	[4]*	[5]**	This work**
Technology (μm)	0.18	0.18	0.13	0.13	0.13
Supply Voltage (V)	1 (1)	0.9	1.2	1	A: 0.8, D: 0.6
Sampling Rate, f_s (kS/s)	200 (100)	200	1000	100	25
Input Frequency, f_{in} (kHz)	100 (50)	100	101	48.5	12.4
ENOB (@f_{in})	7.96 (10.55)	7.44	8.39	9.17	7.81
Power Dissipation (μW)	19 (25)	2.47	150	1	0.35
FoM† (fJ/conversion-step)	381 (166)	65	437‡	17	62

$\star$Measurement results. $\star\star$ Simulation results. $\dagger$ FoM = $P_{diss}/(2 \cdot f_{in} \cdot 2^{ENOB})$
$\ddagger$ FoM = $P_{diss}/(f_s \cdot 2^{ENOB})$

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